## 1. Business Understanding

Weather prediction is important in industries such as agriculture, aviation, and transportation. Despite this, many existing systems are complex and not always practical for quick decision-making.

The aim of this project was to develop a machine learning system capable of predicting weather conditions using atmospheric data, and to present these predictions through a simple web application. The system allows users to input atmospheric values and receive predictions for temperature, rain, snow, and visibility in real time. The focus was on building a solution that is both accurate and easy to use.

## 2. Data Understanding

The dataset used in this project was obtained from Kaggle and is based on Environment and Climate Change Canada (ECCC) data recorded at Toronto Pearson Airport for the year 2012. The dataset contains 8,784 rows, representing hourly observations across a leap year (366 days).

The dataset includes features such as:

- Temperature
- Humidity
- Wind speed
- Pressure
- Visibility
- Weather description (categorical)

After preprocessing and feature engineering, the dataset expanded to include additional features such as time-based variables and encoded weather conditions. The dataset supports both regression and classification tasks. Temperature is treated as a regression problem, while rain, snow, and visibility are treated as classification problems.

During initial exploration, seasonal trends in temperature were observed, along with relationships such as high humidity being associated with rainfall. A key challenge identified was class imbalance, particularly for snow and visibility conditions.

## 3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to better understand the dataset before building the models.

The distributions of variables such as temperature, humidity, wind speed, and pressure were checked to see how the values are spread and to identify any unusual values.

- Temperature showed strong seasonal variation

- Humidity and pressure stayed mostly within expected ranges
- Rain was often associated with high humidity
- Snow occurred mostly at lower temperatures

Time-based patterns were also seen, where certain weather conditions happen more often at specific times of the day or during certain months.

Outliers were identified in variables such as wind speed and pressure, which supported the decision to apply clipping during preprocessing.

Overall, the EDA confirmed that the dataset contained meaningful patterns suitable for machine learning modelling.

## 4. Data Preparation

Data preprocessing was performed to ensure the dataset was suitable for modelling. Missing values and inconsistencies were handled, and categorical weather descriptions were transformed using one-hot encoding through a MultiLabelBinarizer approach.

Feature engineering included splitting the date and time into hour, day, month, and day_of_week variables to capture patterns. A new binary feature, high_visibility, was created to indicate when visibility reached 48.3 km, which represents the sensor's maximum recorded value.

Outliers were handled using statistical methods. Wind speed was clipped at approximately 36.5 km/h based on the interquartile range, while pressure was also clipped using IQR bounds. Humidity values were constrained within the valid range of 0 to 100 percent. Visibility values were not removed, as they were instead represented through the high_visibility feature.

All numerical features were standardised using a StandardScaler to ensure consistent scaling. This transformation ensures that no single feature dominates the model due to differences in scale.

The scaling process follows the formula:
scaled value = (raw value − mean) / standard deviation

The scaler itself was not saved as a file; instead, the mean and standard deviation values were extracted and later hardcoded into the application.

## 5. Modelling

The dataset was split into training and testing sets using a time-based approach rather than a random split. The first 80% of the data was used for training, and the remaining 20% was

used for testing. This approach prevents data leakage and better reflects real-world forecasting scenarios where future data should not influence past predictions.

Different machine learning models were developed depending on the prediction task. Temperature was predicted using a Gradient Boosting Regressor, while rain, snow, and visibility were predicted using Gradient Boosting classification models. Wind speed and pressure were predicted using HistGradientBoosting Regressors, which were selected due to their similar performance to Random Forest models but significantly smaller file sizes, making them more suitable for deployment.

Gradient Boosting models were chosen because they handle non-linear relationships effectively and perform well on structured datasets with both numerical and categorical features.

The temperature model achieved a mean absolute error of approximately 0.077 in scaled units, indicating high accuracy. The rain model achieved around 97% accuracy with strong recall, making it the best-performing model. The snow model achieved approximately 94% accuracy but had lower recall, meaning some snow events were missed. The visibility model achieved around 89% accuracy, although this result is influenced by class imbalance.

## 6. Evaluation

Model performance was evaluated using metrics such as accuracy, precision, recall, and mean absolute error.

The rain model provided the most reliable results, while the snow model showed good performance but missed some cases. The visibility model appeared accurate but was affected by the imbalance in the dataset.

Overall, the models were able to capture key patterns in the data and provide useful predictions, although some limitations remain.

## 7. Web Application Development and Deployment

The interactive web application was developed using Dash, which allows machine learning models built in Python to be integrated directly into an interactive web interface.

The application was designed with simplicity in mind, allowing users to input atmospheric values and receive predictions instantly.

When a user submits inputs, the application applies the same preprocessing steps used during training, ensuring consistency. The processed inputs are then passed to the trained models, and the predictions are displayed clearly on the interface.

For deployment, the trained models were saved and included in the application environment. A requirements file was created to manage dependencies, and the project was pushed to GitHub. The application was then deployed using Render, where it was configured to run directly from the repository. After installation of dependencies and testing, the application was made publicly accessible online.

## 8. Conclusion & Recommendations

This project developed a machine learning system to predict weather conditions using historical atmospheric data.

The models performed well overall, especially for temperature and rainfall prediction. However, some limitations were identified, such as class imbalance and the use of data from only one location and one year.

Based on the results, the following recommendations are proposed:

1. Use data from multiple years and locations to improve generalisation
2. Address class imbalance to improve prediction of rare events
3. Save preprocessing steps properly to ensure consistency
4. Expand the application to include more user inputs

Overall, the project shows how machine learning can be used for weather prediction and how these models can be applied in a simple web application.

## 9. Continuous Improvement

Improvements included simplifying the user interface, improving input validation, and making prediction outputs clearer.

Further improvements will continue as additional feedback is collected and new insights are gained.

## 10. Reflection and Future Work

This project provided practical experience in building a complete machine learning pipeline, from data preparation to deployment. One of the main challenges encountered was handling class imbalance, which affected the prediction of less frequent events such as snow. Integrating the models into a live web application also required careful handling of preprocessing and consistency.

Future work could involve incorporating more advanced models, integrating real-time weather data, and improving the accuracy of predictions for rare events. Enhancing the usability and design of the web application is also an important area for further development.